EASI estimation alongside clinical targets and serum trajectories in a Chinese atopic dermatitis cohort

Abstract

Background: A practical Eczema Area and Severity Index (EASI) estimate should remain interpretable across treatment and follow-up. We developed E-EASI, an estimated EASI combining affected skin area and serum measurements, within a Chinese treat-to-target cohort.

Methods: We examined clinical outcomes and all 23 serum markers under dupilumab and two initial abrocitinib doses, adjusting treatment comparisons for measured baseline differences. E-EASI used nominal-visit-matched affected body surface area (BSA), CCL17, CCL26 and IL-5. Development used baseline/week-12 observations with nested patient-level validation, followed by week-24 application. Individual estimates, within-person changes and EASI75 identification were evaluated separately, with matched input ablation and IgE missing-data sensitivity analyses.

Results: Clinical comparisons included 163 patients; EASI estimation included 316 visits from 160 patients. Both abrocitinib groups had lower week-2 itch scores than dupilumab after adjustment; later differences persisted. Clinical control improved alongside treatment-dependent molecular changes. The selected four-input equation required no treatment label or previous score; performance estimates describe the nested development procedure. Mean absolute error (MAE) at baseline/weeks 12/24 was 3.19/1.30/1.19, with Spearman correlations of 0.918/0.927/0.904. Follow-up errors were within two points in 84.3% and 82.0% of predictions. Matched ablation attributed most accuracy to BSA. Current itch modestly improved hybrid follow-up predictions; molecular-only extensions remained inaccurate. EASI75 sensitivity at weeks 12/24 was 76.5%/86.1%, with specificity of 98.6%/96.2%.

Conclusions: E-EASI gave close follow-up estimates and tracked changes within this cohort. Molecular added accuracy was not established, and baseline severe disease remained the main accuracy boundary. The hybrid measure is reproducible and requires independent evaluation across the severity range before clinical adoption.

Keywords: atopic dermatitis; biomarkers; EASI; longitudinal estimation; treat-to-target.

1 Introduction

Atopic dermatitis treatment must improve visible skin disease, itch and the burden of living with eczema. Treat-to-target care links repeated assessment of these outcomes to treatment review.[1, 2] A useful severity estimate should express an individual patient's current condition and remain interpretable as treatment changes. This is especially relevant when rapid symptom relief precedes the later control of skin disease.

Existing frameworks differ in their timing and required depth of response. International recommendations link patient global assessment to disease-domain targets, whereas Canadian recommendations require both clinician-reported and patient-reported improvement. Chinese guidance includes earlier visits to assess itch relief.[2–5] The AHEAD approach defines minimal disease activity through optimal targets selected with the patient; the International Eczema Council distinguishes low and very low disease activity from duration-defined control and remission.[6, 7] These approaches preserve several clinical perspectives. They do not establish a blood-based measure that can recover the same clinical information.

Serum markers offer a link between clinical activity and inflammation. The p-EASI approach combines CCL17, IL-22 and soluble IL-2 receptor, with different coefficients before and during treatment.[8] Subsequent studies reported Spearman correlations of 0.71 during cyclosporin treatment and 0.67 during dupilumab treatment.[9, 10] These studies established a practical precedent for converting blood measurements into Eczema Area and Severity Index (EASI) units. Applying such a calculation in another setting also requires agreement with individual scores, compatible assays and evaluation across the severity range.[11, 12]

EASI combines the extent of affected skin with lesion intensity.[13, 14] A small-input equation combining affected body surface area (BSA) and serum measurements could offer a common calculation without separate rules for each drug or treatment phase. BSA supplies the clinical extent component; the serum panel supplies molecular information. Whether the latter improves accuracy must be tested directly. We use E-EASI, meaning estimated EASI, for this clinical–molecular research measure. It is intended to estimate recorded EASI at the visit; measured itch, quality of life and safety remain complementary assessments.

We investigated these questions in a Chinese cohort receiving dupilumab or two initial abrocitinib doses. We characterized clinical outcomes within the local target framework and followed all 23 measured serum markers, accounting for baseline differences and later treatment changes. We then developed a common BSA–serum equation and evaluated individual estimates, changes and response classification using patient-level validation. Matched input comparisons and transfer of molecular-only formulas to a published Chinese dataset tested what the molecular measurements contributed.

2 Methods

2.1 Study population and clinical assessment

We analysed longitudinal records from a Chinese observational treat-to-target project at CAMS and Huashan. The supplied protocol specified Hanifin–Rajka diagnosis and eligibility for systemic treatment. Actual screening implementation and recruitment dates were not recoverable from a verified screening register. Systemic treatment was clinician directed. Initial regimens were dupilumab, abrocitinib 100 mg or abrocitinib 200 mg; later dose, interval and regimen changes were recorded separately. Clinical comparisons included patients aged at least 12 years with an unambiguous initial regimen. Adult Dermatology Life Quality Index (DLQI) analyses required age at least 18.

Clinical outcomes were the recorded EASI, peak pruritus numerical rating scale (PP-NRS), Investigator Global Assessment (IGA), DLQI, BSA and SCORAD, with IgE and eosinophil measurements. Reporting visits were W0/W2/W4/W12/W24 for PP-NRS, W0/W12/W24 for IgE and W0/W4/W12/W24 for other measures. Records were linked by patient and nominal visit. The supplied protocol describes a 24-hour PP-NRS recall window but its actual implementation, the exact IGA/questionnaire versions and calendar-day clinical/serum matching require confirmation. Source discrepancies were audited without substituting imputed clinical outcomes.

2.2 Clinical framework, targets and treatment pathways

Project materials describe evidence review, input from physicians, nurses and patients, two questionnaire rounds with feedback, and steering-group review. The planned agreement criterion was at least 75%. They report 49 returned questionnaires, without original ballots or round-specific agreement results. We transcribed the local framework and operationalized its targets on recorded scores (Fig. 1). Its exact approved and prospectively implemented version remains unverified; this analysis does not validate a new consensus or the effectiveness of the management strategy. A literature comparison distinguishes this framework from international and Chinese recommendations, AHEAD minimal disease activity and the IEC disease-control states (Supplementary Table S13).[2–4, 6, 7]

The week-2 target was PP-NRS ≤ 4. At week 4, the physician target was EASI50 or an IGA decrease of at least two points, and the patient target was PP-NRS ≤ 4 or adult DLQI ≤ 10. At week 12, these were EASI75 or IGA 0/1, and PP-NRS 0/1 or adult DLQI ≤ 4. Week 24 additionally allowed EASI ≤ 5 in the physician domain. Both domains were required for the combined target. Individual reporting thresholds additionally included EASI ≤ 3; this did not change the stage-specific composite. These investigator-defined targets were kept distinct from newer low-disease-activity and remission definitions.[7]

Each rate used its own evaluable denominator and Wilson 95% confidence interval. EASI percentage responses required positive observed baseline EASI. A PP-NRS decrease of at least four points required observed baseline PP-NRS ≥ 4 and an observed follow-up score; lower or missing baseline values were excluded from that endpoint. Missing components remained unknown unless another observed component determined the logical result. Below age 18, adult DLQI was treated as unknown: a positive itch component establishes the patient target, whereas a negative itch component alone leaves it unknown. Group comparisons retained initial treatment after subsequent changes. Pathway summaries described week-12 response and subsequent documented treatment changes as proportions of each complete initial group. Exploratory week-24 response summaries and missing-outcome bounds remain in the source tables. Reduction was not reassigned to an earlier visit; uncertain assessment/decision timing was retained. These descriptive pathways address treatment changes without treating a response-selected maintenance group as randomized.[15]

2.3 Serum measurements

Source laboratory reports described Luminex and MSD serum assays, including the LXSAHM-13 panel on Luminex-200 and K15067M-1/K15067M-2 kits on SQ120, using 50-μL serum and twofold dilution. All 23 measured markers were included. Analyses used positive numeric integration-export concentrations recorded in pg/mL, transformed to $\log_2$ once without an additional dilution correction. The twofold laboratory dilution

describes sample preparation; its correction in the export and validated quantification limits require source confirmation. Censoring strings and nonpositive values were not treated as measured positive concentrations. An IL-5 calibration-bound sensitivity analysis and detailed assay provenance are provided in the Supplementary Methods; software calibration limits were distinguished from validated quantification limits.

Molecular changes were calculated within patient before summarization. Each two-visit analysis used all available pairs for its interval: W0–W12, W12–W24 or W0–W24. It did not require a measurement at the third visit. The main three-visit display instead required all three measurements within each marker and initial treatment. It expressed W12/W24 as mean within-person $\log_2$ changes from W0, keeping the same patients at both follow-ups. Both analyses were unadjusted within-group summaries; covariate-adjusted between-treatment contrasts were reported separately.

2.4 EASI estimation and validation

Eligible model visits required age at least 12, observed EASI and BSA, and positive observed CCL17, CCL26 and IL-5. The three markers were retained as a biologically motivated compact candidate with good assay coverage, rather than selected as the best three markers by a single automated search. The earlier candidate formation was exploratory and was not repeated within the present validation. Development used W0/W12; W24 was excluded from fitting and selection. As W24 had been examined in previous project analyses, it represents within-cohort temporal application, not untouched external validation.

A repeated five-fold patient-level outer validation with three-fold inner validation selected among three common structures and ridge penalties. All records from a patient stayed in one outer fold, preprocessing used training data only, and patients received equal total fitting weight. The candidate structures were linear, quadratic in BSA, or an area interaction in which BSA multiplied a linear molecular term. Structure and ridge penalty were selected within the patient-grouped inner folds. The final coefficient vector was then fitted to all development observations; the equation is reported in Results.

Primary assessment used MAE with patient bootstrap intervals conditional on the saved held-out predictions; these intervals did not repeat model selection or fitting. This internal validation does not include the earlier selection of the marker panel. Additional measures were RMSE, signed error, Spearman correlation, Lin's concordance correlation and proportions within one or two EASI points. Repetition-specific metrics were calculated before averaging. Severity strata and centre/treatment holdouts assessed where performance varied. A separate refitting sensitivity removed development visits with complete regional scores whose sum differed from recorded total EASI by more than 0.15 points. It retained the original patient folds and outcomes and repeated inner selection; it tested exclusion robustness without correcting either clinical reference. Monitoring used estimated and observed changes within the same patient, repetition and fold; response classification used the corresponding estimated or observed scores at both ends of the interval. Matched controls included BSA alone and the three markers alone. Primary paired comparisons used the nested procedure; fixed-structure comparisons were labelled separately. Published p-EASI coefficients were applied unchanged as a source-table sensitivity analysis, using local IL2RA as an unbridged soluble IL-2 receptor proxy and retaining raw and matched-clipping results. Different inputs and unverified assay equivalence preclude a model-superiority claim.[8–10] PP-NRS, adult DLQI and recorded IGA were fitted independently with target-appropriate regression or ordinal models using the same restricted inputs.

Post hoc input ablation compared serum-only, BSA-only and hybrid models on identical visits and patient folds. A current-itch comparison retained 151/89/75 W0/W12/W24 observations; adult DLQI was tested separately on 105/58/44 matched adult observations. Every comparator was refitted within its comparison population. Structure and penalty selection remained inside the W0/W12 training folds. Fixed-area and single-marker controls distinguished molecular inclusion from changes in model structure. Paired patient bootstrap differences were exploratory and not adjusted across the multiple input comparisons; full specifications appear in the Supplementary Methods.

Exploratory molecular-only analyses retained patient-level nested validation while testing nonlinear, paired-change and auxiliary-target structures (Supplementary Table S14). These analyses did not alter the primary formula. Separately, existing molecular-only equations and published p-EASI were applied without refitting to the same eligible observations in OMIX002850, with identical 0–72 prediction bounds and patient-paired bootstrap comparisons. Because BSA was unavailable, this comparison did not evaluate E-EASI. Prior dataset exploration, possible participant overlap and assay differences precluded independent-validation claims (Supplementary Table S15).

2.5 Statistical analysis

Patients were the independent units. Baseline continuous variables were summarized by the median and interquartile range, using linear interpolation at sorted rank $1+(n-1)p$. This matches Excel PERCENTILE.INC; changing the quartile convention changes summaries without changing patient records. Main treatment-specific trajectories

show observed medians and interquartile ranges; adjusted mean comparisons are reported separately. Continuous treatment comparisons used baseline-outcome adjustment, treatment-by-visit terms, age, sex, BMI and centre, generalized overlap weights and patient-cluster robust covariance. Weights target overlap in baseline treatment probabilities. Continuous-outcome intervals condition on the estimated baseline weights; they do not propagate propensity-estimation uncertainty. Reported outcome contrasts are baseline-weighted associations among patients with that endpoint observed. Selective endpoint observation can therefore change the represented population and its covariate balance.[16] Weighted target contrasts used patient bootstrap intervals with propensity refitting. A W2 sensitivity additionally included baseline PP-NRS, comparing models on the same baseline-itch-complete cohort. Endpoint-specific balance, effective sample sizes and both weighted and unweighted missing-outcome bounds were retained; weighting did not correct informative follow-up loss. IgE/eosinophil models used $\log(1 + x)$ and back-transformation. The analyses estimate adjusted observational associations.

Laboratory display models used every available primary-visit measurement without requiring complete follow-up. Maximum-likelihood random-intercept models of $\log(1 + \text{IgE})$ or $\log(1 + \text{EOS})$ included initial-treatment-by-categorical-visit profiles, $\log(1 + \text{baseline EASI})$, age, sex, BMI and centre. Treatment-specific baseline laboratory means were freely estimated. Fixed-effect predictions were standardized to baseline covariates and back-transformed as shifted geometric means; changes were ratios of (1+outcome) to each treatment's modelled baseline. Model-based 95% intervals used fixed-effect covariance. W12/W24 between-treatment change contrasts used Holm correction across six comparisons per outcome and analysis. CAMS-only and centre-by-visit fits assessed cross-centre assumptions because Huashan had no dupi follow-up IgE. Earlier patient-wide multiple imputation, with 400 retained datasets per method, and missing-not-at-random scenarios remained separate sensitivities. No observed value was replaced.[17, 18]

All W4 IgE and dupi W4 eosinophil measurements were unobserved. Hollow W4 display points interpolated fitted W0/W12 $\log(1 + \text{outcome})$ means by elapsed time, with conditional intervals from endpoint covariance. They were illustrative temporal assumptions, not identified W4 measurements or treatment-test inputs. Alternative temporal shapes and the earlier paired-patient eosinophil sensitivity were retained. No estimated clinical score entered prediction validation.

Unadjusted target comparisons used conditional Monte Carlo three-group tests and pairwise Fisher tests; the three pairwise contrasts within each endpoint/visit formed a Holm family. The fixed-three-visit molecular display used paired sign-flip tests and Benjamini–Hochberg adjustment across 23 markers, three treatments and two follow-up contrasts: 138 planned hypotheses, 124 estimable. The all-available-pair analysis used the same type of paired test, with BH correction across 193 estimable hypotheses among 207 planned marker–treatment–window comparisons. Its populations and multiplicity family differed from the three-visit analysis; its stars were not transferred to the main figure. Between-treatment molecular changes additionally adjusted for starting concentration and clinical baseline variables. Estimates based on fewer than five pairs were descriptive. Balance, effective sample size and common support were examined. Exact tests, numerical settings, model diagnostics, regional analyses and additional targets are detailed in the Supplementary Methods. Analyses used Python 3.9.6; dependency versions, code, data-access conditions and reproduction logs accompany the package. Reporting follows relevant STROBE and TRIPOD+AI domains.[19, 20]

3 Results

3.1 Clinical framework and analysis populations

The local framework progressed from evidence review and stakeholder input to two questionnaire rounds and steering-group review. It combined clinical-sign and patient-experience targets at W4, W12 and W24; W2 assessed itch alone (Fig. 1). Its timing and target definitions are compared with other frameworks in Supplementary Table S13. E-EASI was evaluated as a parallel research measure and did not determine treatment decisions.

The clinical comparison included 163 patients: 65 started dupilumab, 56 abrocitinib 100 mg and 42 abrocitinib 200 mg. Median baseline EASI was 15.20, 11.70 and 19.35, respectively; age, sex and BMI also differed (Table 1). Clinical treatment comparisons and linked-serum modelling used separate eligibility rules, detailed in Supplementary Fig. S1.

3.2 Clinical control improved under the three initial treatment strategies

Observed EASI, PP-NRS and adult DLQI decreased from W0 to W12 and W24; the component target rates describe the corresponding clinical control (Fig. 2). At W2, PP-NRS was at most 4 in 15/27 evaluable dupilumab recipients, 34/37 receiving initial abrocitinib 100 mg and 33/36 receiving initial abrocitinib 200 mg. The adjusted 100 mg–dupilumab difference was 41.7 percentage points (95% CI, 15.1–69.4; Holm-adjusted $P = 0.0095$). Adding baseline itch gave a similar difference of 41.4 points (13.0–66.8). W2 itch was unavailable in 38/65, 19/56 and

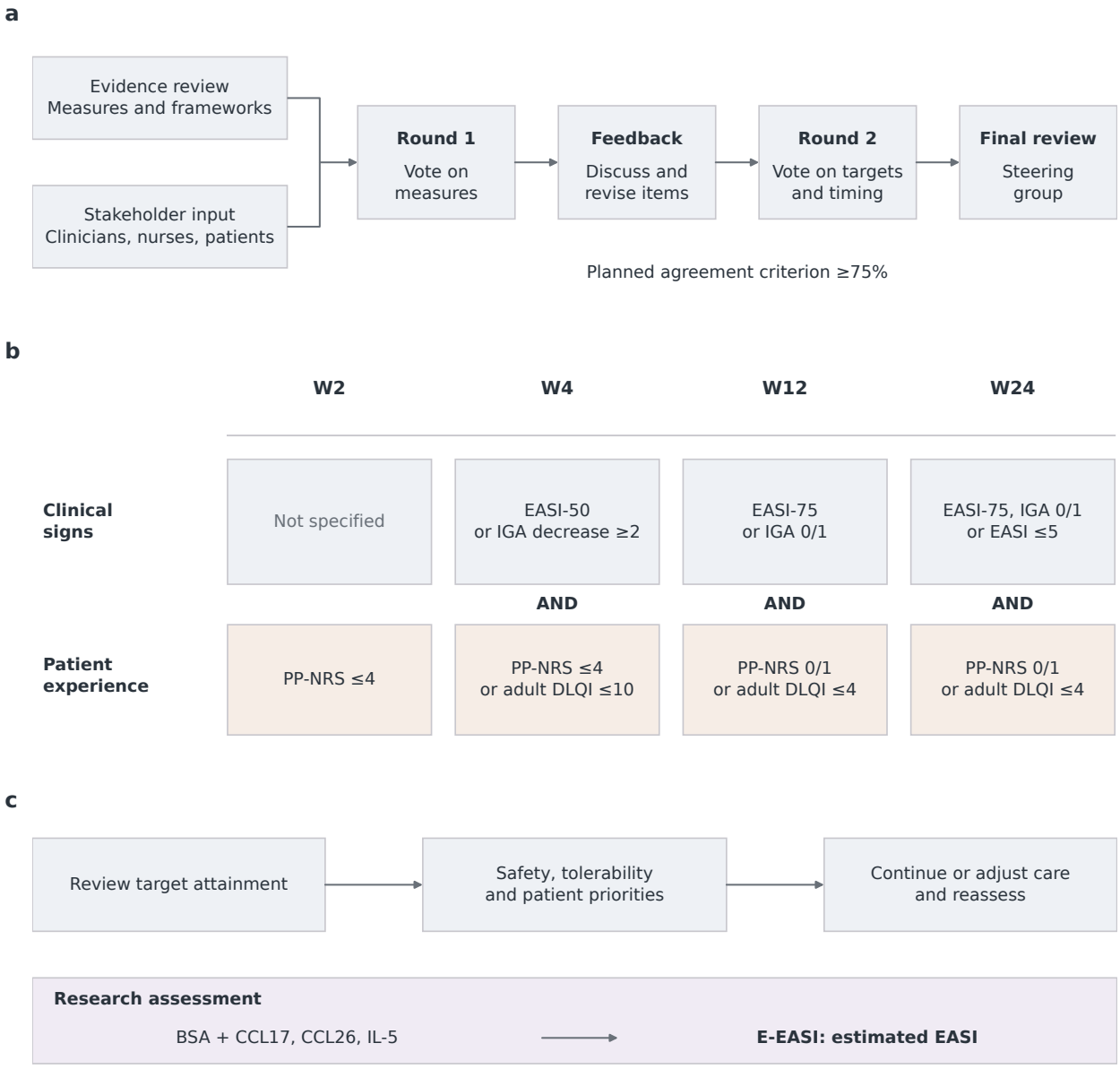


Figure 1: **Development, clinical targets and the E-EASI research assessment.** a, The project materials describe evidence review, clinician/nurse/patient input, two questionnaire rounds, feedback and steering-group review. The planned agreement threshold was 75%; 49 returned questionnaires were reported. Original ballots and round-specific participation and agreement are unavailable, so the diagram reconstructs the reported process rather than verifying consensus. b, Operational targets for W2/W4/W12/W24. At W2 only PP-NRS $\leq$4 is required. From W4 onward at least one criterion in each domain is required; "or" is within a domain and "AND" joins domains. EASI-50/75 means at least 50%/75% improvement from positive pretreatment EASI; IGA decrease $\geq$2 is relative to baseline. IGA 0/1 and PP-NRS 0/1 mean recorded values 0 or 1. DLQI is analysed in adults aged $\geq$18. Missing components remain unknown unless another observed component resolves the domain. The recorded IGA instrument and implemented framework version require investigator confirmation. Longer-term planned targets and other published frameworks are compared in Table S13. c, Target review accompanies safety and patient priorities. E-EASI is a separate, post hoc hybrid clinical–molecular research estimate; it did not determine treatment changes. This observational cohort does not establish that the framework or the research estimate improves management.

Table 1: Baseline characteristics of the clinical comparison population.

	Dupilumab (n=65)	Abrocitinib 100 mg (n=56)	Abrocitinib 200 mg (n=42)
Age, years	54.0 (29.0–71.0)	37.5 (22.0–52.0)	24.5 (19.0–46.8)
BMI, kg/m^2	23.0 (21.0–26.0)	22.4 (20.2–24.3)	23.6 (21.4–26.0)
Male sex, n/N (%)	36/65 (55.4)	27/56 (48.2)	35/42 (83.3)
EASI	15.2 (11.9–21.6)	11.7 (7.7–17.3)	19.4 (9.6–27.8)
PP-NRS	8.0 (7.0–9.0)	7.0 (5.5–8.0)	7.0 (5.0–9.8)
Available n	65	55	42
IGA, recorded grade	4.0 (3.0–4.0)	3.0 (3.0–4.0)	4.0 (3.2–4.8)
Adult DLQI	15.0 (10.5–21.0)	10.5 (7.0–15.2)	15.0 (10.2–21.8)
Available n	47	32	34
BSA, %	23.0 (17.0–37.0)	16.0 (10.0–25.2)	25.0 (15.5–36.2)
SCORAD	50.0 (43.2–59.4)	48.1 (40.4–54.2)	58.1 (48.7–72.6)
Available n	65	44	41
Eosinophils, recorded units	0.2 (0.1–0.4)	0.3 (0.1–0.6)	0.5 (0.3–0.7)
Available n	34	51	38
IgE, recorded units	200.0 (57.2–741.2)	214.7 (71.0–626.0)	700.0 (57.6–5000.0)
Available n	62	41	37

Values are median (interquartile range), unless stated otherwise. Quartiles use linear interpolation at rank $1 + (n - 1)p$ in the sorted observations (equivalent to Excel PERCENTILE.INC). Adult DLQI is restricted to age ≥ 18. IGA is retained on the recorded scale pending instrument confirmation. The column-header n applies unless an available n is shown. Missing values are not filled.

6/42 patients, respectively, leaving these associations sensitive to follow-up selection. Weighted worst-case bounds permitted reversal of the W2 contrast (Supplementary Methods).

Continuous itch comparisons also favoured both initial abrocitinib groups at W2. Adjusted PP-NRS was lower than with dupilumab by 2.14 points for 100 mg (95% CI, 0.82–3.46) and 2.24 for 200 mg (0.83–3.65; both Holm-adjusted $P = 0.0049$). Differences remained at W24: dupilumab-minus-abrocitinib contrasts were 1.62/1.60 PP-NRS points and 2.98/3.32 EASI points for 100/200 mg, respectively, with all four Holm-adjusted $P < 0.015$. The estimated itch gaps were 0.52 and 0.64 points smaller at W24 than W2, but direct intervals for these changes included zero (-1.98 to 0.93 and -2.31 to 1.03 for later-minus-early differences). These data support an early itch association, without establishing convergence or equivalence of later efficacy. Complete estimates and intervals are supplied in the clinical contrast tables.

Recorded IGA distributions shifted towards lower grades (Fig. 3). At W24, the combined physician- and patient-reported target was reached by 26/36 (72.2%), 22/24 (91.7%) and 20/23 (87.0%), respectively (Fig. 2o). Counting all unknown outcomes as failures or successes gave initial-cohort bounds of 40.0–84.6%, 39.3–96.4% and 47.6–92.9%. Later management included continuing and reducing treatment. Pathway source tables retain all initial patients and distinguish W12 response, subsequent reduction and unavailable management records. They describe management after response, without estimating a randomized dose-reduction effect.

Removing the IGA component from the reconstructed framework changed combined-target success to failure in 4/90, 6/112 and 0/83 jointly evaluable patients at W4/W12/W24, respectively (Supplementary Table S10). This definition sensitivity preserved the W24 combined classifications; it did not validate the implemented IGA scale.

The lower unadjusted EASI in the 100 mg group did not persist as a clear dose-group difference after baseline adjustment. The adjusted 100-minus-200 mg difference was 0.62 points at W12 (95% CI, -2.08 to 3.33) and 0.34 at W24 (-1.10 to 1.77). Weighting improved baseline balance, although endpoint-specific residual imbalance remained (Supplementary Fig. S2). Three W0 regimen entries conflicted with the initial-dose label and remain unresolved. Safety documentation through W24 was present for 27/65, 20/56 and 19/42 patients; these counts do not establish event incidence or treatment attribution (Supplementary Table S3).

BSA, SCORAD and laboratory measures appear in Supplementary Figs. S3–S4; all target contrasts appear in Supplementary Figs. S7–S10. The all-available IgE model retained 258 measured values from 147 patients, compared with 30 patients with all three visits. Its W24-to-baseline ratios of $(1 + \text{IgE})$ were 0.38 (95% CI, 0.23–0.61) for dupi, 0.70 (0.41–1.17) for abro 100 mg and 0.85 (0.51–1.43) for abro 200 mg (Supplementary Fig. S5; Supplementary Table S17). The 100-versus-200 mg ratio of baseline-relative change was 0.81 (0.39–1.69; Holm-adjusted $P = 1.00$). CAMS-only and centre-by-visit sensitivities likewise did not establish a dose difference. Thus the ordering of these curves does not show a superior IgE response with 100 mg. Earlier donor and Gaussian imputation results remained sensitive to missing-data assumptions for abro 200 mg (Supplementary Table S8).

Huashan contributed 20/5/4 observed IgE values at W0/W12/W24, but none of its dupi follow-up values. Pooled trajectories therefore required a shared centre-pattern assumption; the CAMS-only curves supplied a directly supported follow-up sensitivity. Eosinophil modelling used 320 measurements from 138 patients. No W4 IgE or dupi W4 eosinophil value was measured. Hollow points in the fitted trajectories illustrate a smooth W0–W12 interpolation, not restored laboratory observations (Supplementary Figs. S4b,d and S5). The fitted dupi W4 eosinophil level was 0.310 (conditional 95% CI, 0.243–0.381); its interval does not account for an unobserved

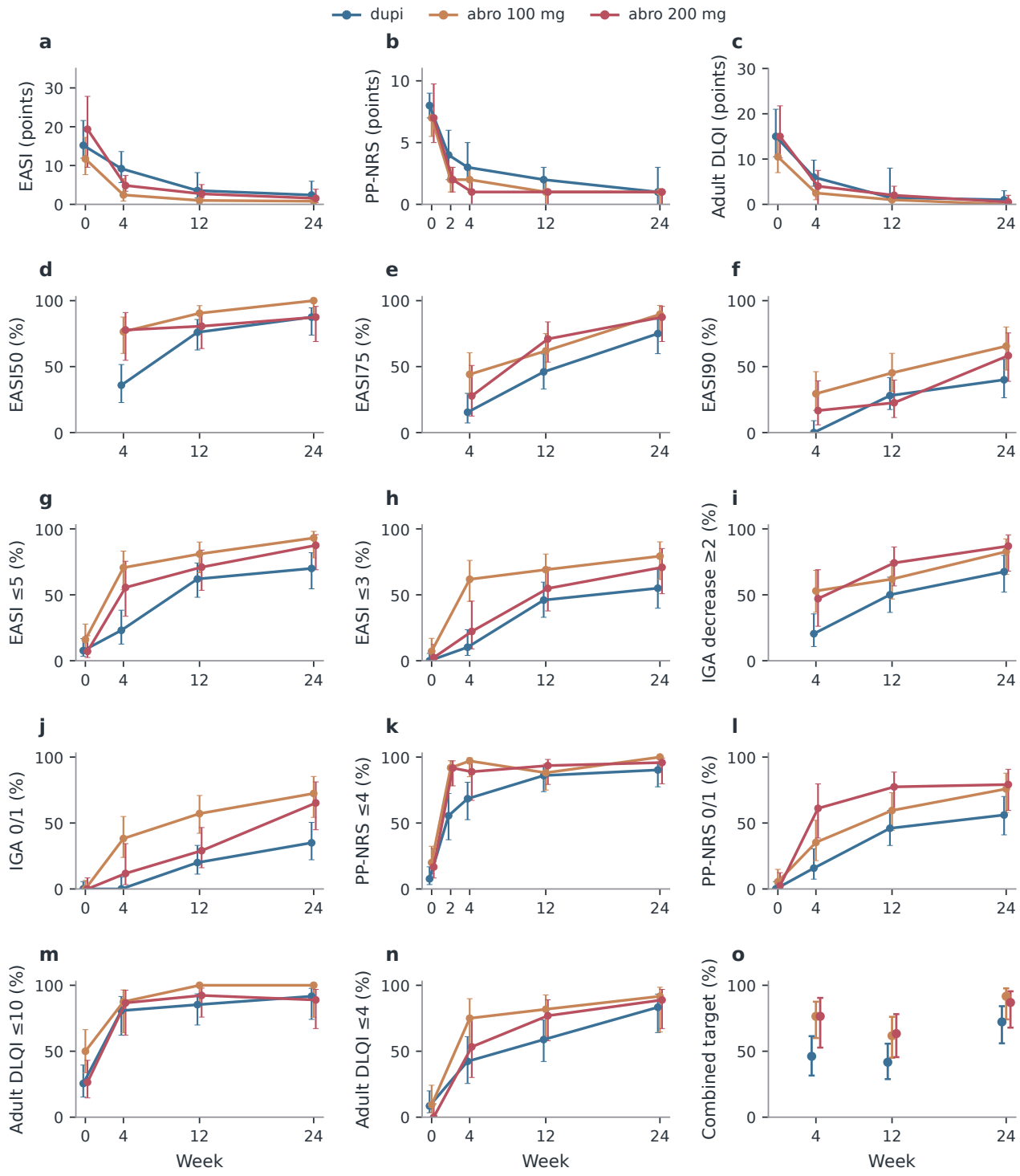


Figure 2: **Observed clinical activity and target attainment by initial treatment.** a–c, Observed medians and interquartile ranges for EASI, PP-NRS and adult DLQI. d–n, Clinical threshold attainment, with pointwise Wilson 95% confidence intervals. Each visit uses its available observations, not a fixed patient set. Group positions are slightly offset for legibility. Improvement thresholds require observed baseline and follow-up; adult DLQI requires age ≥18. o, Combined physician-and-patient targets from Fig. 1, with Wilson intervals. These stage-specific criteria differ, so points are not connected. Colour denotes initial treatment throughout; subsequent treatment changes do not reassign groups. No outcomes are imputed. All target numerators and denominators are in Supplementary Table S1; headline missing-outcome bounds are in Results and headline_missing_outcome_bounds.csv. Baseline-adjusted treatment comparisons are in the supplement. dupi denotes dupilumab; abro denotes abrocitinib. These available-case, observational trajectories do not establish causal treatment differences.

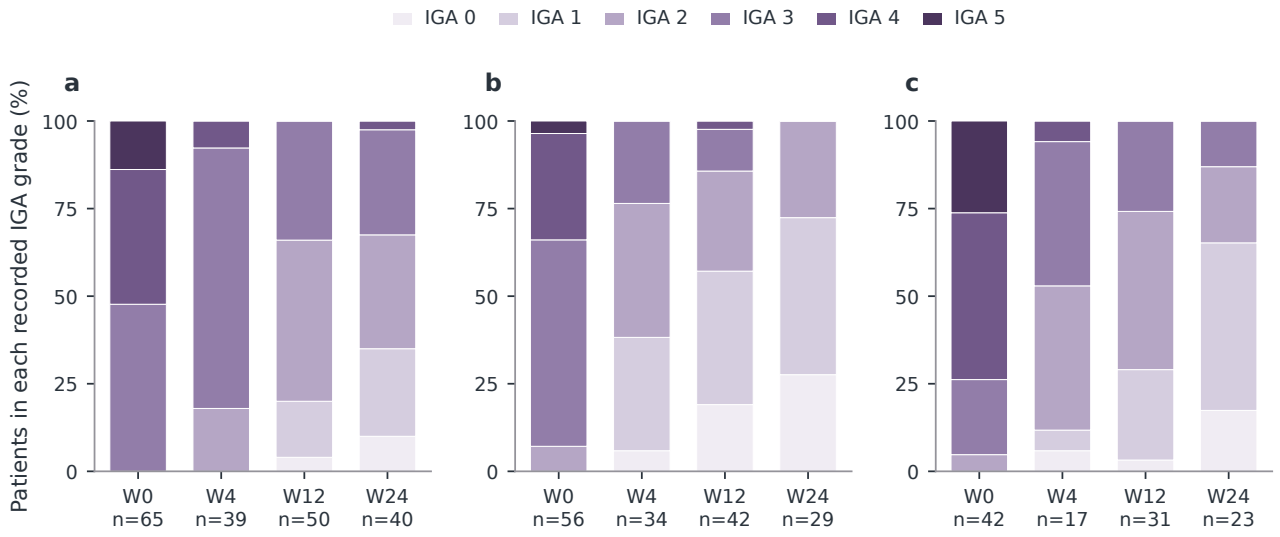


Figure 3: **Recorded IGA distributions during follow-up.** a–c, Distribution of recorded IGA grades within each initial treatment group at W0, W4, W12 and W24. Each bar totals 100% of the patients with an observed grade; n below the bar is that visit's denominator. Colour identifies grade, not treatment. Patient sets may differ between visits. Grade-specific counts and proportions are provided in the source table. The local 0–5 coding is retained; correspondence to a validated IGA instrument has not been confirmed. This figure describes disease activity, not dose-reduction success. Panel key: a, dupi; b, abro 100 mg; c, abro 200 mg.

transient rise. The earlier 15-patient paired-median interpolation, a different population and statistic, remains in Supplementary Fig. S6 as a sensitivity. A post hoc application of the IEC EASI-and-itch thresholds classified 91/123 patients (74.0%) as having low disease activity at W12 and 81/93 (87.1%) at W24. Very low disease activity was present in 50/123 (40.7%) and 54/93 (58.1%), respectively. These proportions use patients with both scores observed at that visit; treatment-specific rates and full-group missing-outcome bounds are given in Supplementary Table S16. They describe single-visit states, not sustained control or remission.

3.3 Serum trajectories differed across initial treatment groups

All 23 measured markers were retained. Among all available W0/W12 pairs, dupi recipients had lower CCL17, CCL18, CCL26, CCL11, IL-36β and IFN-β at W12 (BH q=0.0011–0.0325; Supplementary Fig. S11). Figure 4 instead follows only patients with all three visits available for each marker. In that subset, CCL17 and CCL26 met the W12 significance criterion in dupi; the other four did not. CCL17 remained lower at both follow-ups in dupi and abro 200 mg, as did IL2RA in both abrocitinib groups (all q$<$ 0.05 within the 138-comparison family). These unadjusted within-group findings do not test differences between drugs.

Baseline concentrations are reported directly as medians, quartiles and evaluable counts in Supplementary Table S12. Supplementary Figs. S11–S13 use all available pairs for each interval, with no requirement for a third measurement. Their patient sets and multiplicity family differ from the fixed-three-visit analysis. Baseline-adjusted between-treatment comparisons, including all three pairwise contrasts and dupi versus the observed mixture of abrocitinib doses, appear in Supplementary Figs. S14–S16. For CCL17 over W0–W12, the adjusted dupi-minus-abro 100 mg difference was -1.07 $\log_2$ units (95% CI, -1.67 to -0.48; q=0.0109). Its assay-batch sensitivity estimate was -0.72, with an interval crossing zero (q=0.369), showing that this particular contrast depended on the adjustment specification.

Selected molecular changes accompanied clinical changes. In abro 100 mg over W0–W12, IL-5 change and PP-NRS change had Spearman $\rho = 0.742$ (31 pairs; panel-wide q=0.00156). The declared association family also identified TNF-α and TSLP changes with EASI change over W0–W24. Supplementary Fig. S17 displays all associations meeting the existing panel-wide FDR criterion, across all windows. The complete family, including nonsignificant and non-estimable cells, remains in the source tables; selection for this display is exploratory. Regional disease and baseline molecular associations are brought together in Supplementary Fig. S18.

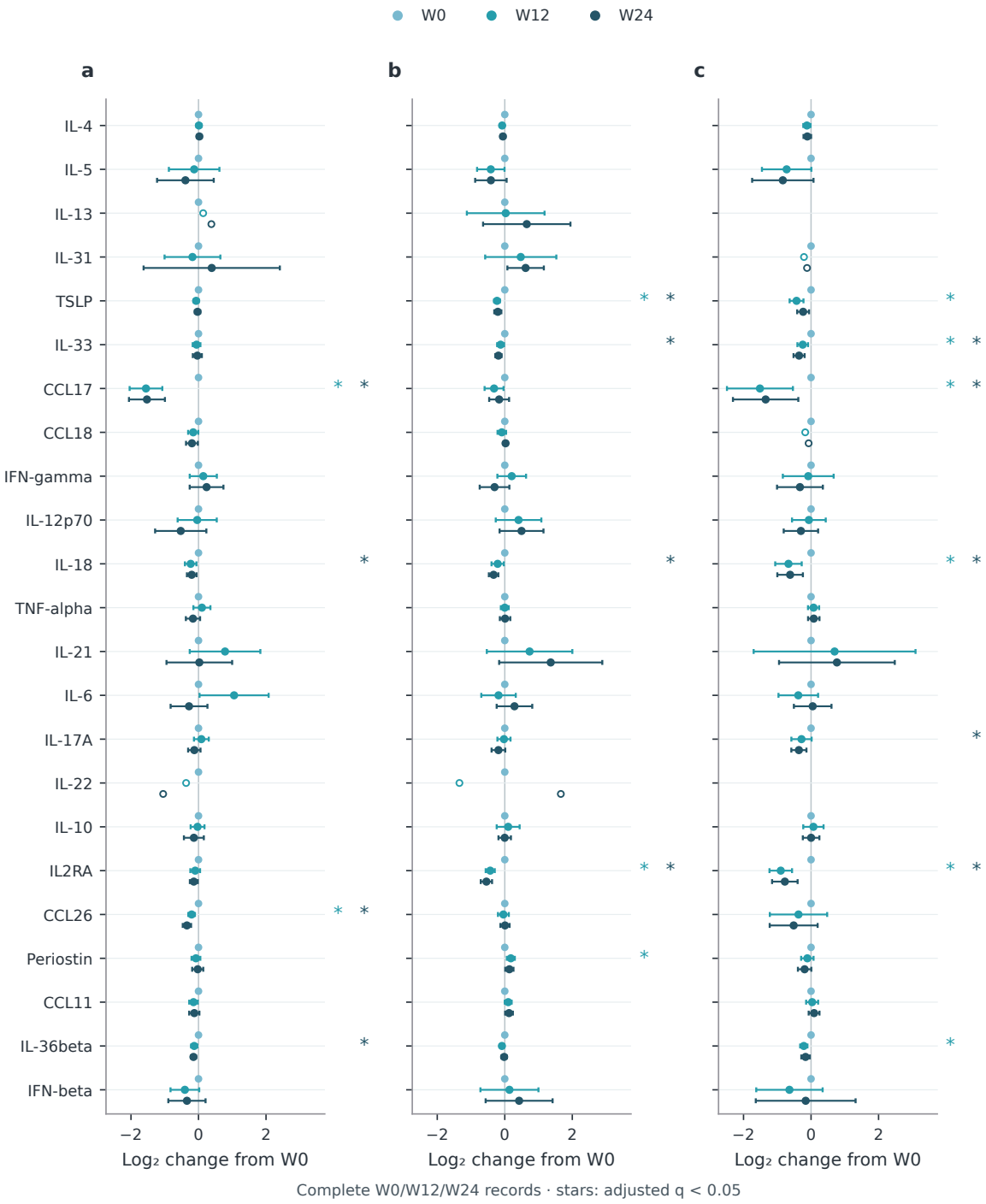


Figure 4: **Molecular trajectories in patients with all three visits.** a–c, Only patients with an available result at all three visits (W0, W12 and W24) for the stated marker and initial treatment are included. The same patients contribute at every point within that row; sets can differ between markers and treatments. These are a subset of the W0/W12 pairs in Supplementary Fig. S11, so the W12 estimates and stars need not match that all-available-pair analysis. W0 is the reference value zero, not zero concentration. Coloured points show mean within-person $\log_2$ changes from W0; horizontal bars are 95% t confidence intervals for $n \geq 5$. A change of -1 means a geometric paired concentration ratio of 0.5. Hollow later-visit points denote $n < 5$ without inference; blank rows have no complete three-visit set. A coloured star beside a marker row refers to that same-colour visit versus W0: *BH $q < 0.05$ across all 138 planned marker-by-treatment-by-contrast tests, using paired sign-flip tests and retaining non-estimable hypotheses in the family. These stars are calculated in the displayed fixed patient sets, not copied from analyses of all available pairs. No baseline covariate adjustment is applied to this descriptive figure. Between-treatment adjusted comparisons and all available-pair windows are reported separately in the supplement. Baseline concentrations are given in Table S12; exact n, changes, intervals and q values are in the source table. Panel key: a, dupi; b, abro 100 mg; c, abro 200 mg.

3.4 One equation estimated follow-up EASI closely across visits

The model dataset contained 316 complete-input visits from 160 patients: 152 at W0, 89 at W12 and 75 at W24. Development used 241 W0/W12 visits from 156 patients. The final equation combined BSA, CCL17, CCL26 and IL-5 with the same calculation before and during treatment. It required no treatment label, visit week or previous score. Observed-versus-estimated score distributions are shown in Supplementary Fig. S19.

The fitted equation multiplied affected area by a serum-based component. The two-decimal display was

$$\widehat{\mathrm{EASI}} \approx \mathrm{clip}_{[0,72]}\Big\{\frac{\mathrm{BSA}}{100}\big[51.27 + 0.98\log_2(\mathrm{CCL17}) + 0.96\log_2(\mathrm{CCL26}) - 0.83\log_2(\mathrm{IL\text{-}5})\big]\Big\}. \tag{1}$$

BSA is entered from 0 to 100 and concentrations use the development export's recorded pg/mL numerics, with the logarithm applied once. Here, clip bounds the result to 0–72, and $\log_2$ is the base-two logarithm. E-EASI denotes this estimated EASI; it is a hybrid clinical–molecular measure, not a separately calibrated blood-only inflammation scale. The calculator retains full-precision parameters; the displayed approximation differed from final-equation predictions by less than 0.08 EASI points in this dataset. Threshold decisions use the precise calculator. Treatment, stage, time and previous scores are not inputs. Reported primary performance evaluates the nested selection procedure; it is not the apparent fit of this final equation.

MAE under nested patient-level validation was 3.19 at W0 (95% CI, 2.75–3.70) and 1.30 at W12 (0.96–1.71); same-cohort W24 temporal application gave 1.19 (0.89–1.53). Spearman correlations were 0.918, 0.927 and 0.904; concordance correlations were 0.910, 0.900 and 0.869 (Table 2). The error distributions (Fig. 5d) showed that 41.8%, 84.3% and 82.0% of predictions were within two EASI points. These describe individual errors, rather than agreement between group means. Mean signed errors at W12/W24 were +0.82/+0.84 points, with calibration slopes of 0.81 (95% CI, 0.66–0.94) and 0.73 (0.59–0.82), indicating residual overestimation and scale miscalibration.

Table 2: Primary performance of the common-model development procedure and matched controls.

	W0	W12	W24
Patients	152	89	75
MAE (95% CI)	3.19 (2.75, 3.70)	1.30 (0.96, 1.71)	1.19 (0.89, 1.53)
BSA-only area MAE	3.12	1.18	1.02
Three-marker-only MAE	7.63	7.57	8.69
Primary minus BSA-only area MAE difference (95% CI)	0.07 (−0.03, 0.17)	0.12 (0.04, 0.20)	0.17 (0.11, 0.24)
RMSE	4.45	2.29	1.87
Mean signed error	−0.59 (−1.27, 0.07)	0.82 (0.43, 1.29)	0.84 (0.50, 1.25)
Spearman correlation	0.918	0.927	0.904
Concordance correlation	0.910	0.900	0.869
Within two points, %	41.8	84.3	82.0
Severity-band agreement, %	83.2	88.2	92.7

W0/W12 use patient-grouped nested validation; W24 is same-cohort temporal application. Structure and penalty selection are internal to each training fold; earlier marker-panel selection is not repeated. Matched controls use the same patients and outer folds. BSA-only area uses the zero-intercept area structure; three-marker-only uses CCL17, CCL26 and IL-5 without BSA. Positive paired MAE differences mean greater primary-procedure error. Metrics are averaged after scoring each repetition. Intervals are 95% patient-bootstrap intervals conditional on saved predictions; selection and fitting are not repeated. MAE, mean absolute error; RMSE, root mean squared error. Positive signed error indicates overestimation. Severity-band agreement uses four EASI bands for observed and predicted values: ≤ 7, > 7 to ≤ 21, > 21 to ≤ 50, and > 50. The single final formula is not independently validated.

Baseline severe disease remained the main accuracy boundary: MAE was 5.38 among 44 patients with EASI > 21, compared with 1.15 and 2.55 in the lower strata. Full severity-stratified estimates remain in the model source tables. Only one W12 observation and no W24 observation had EASI > 21. Centre and treatment holdouts also varied; their complete estimates remain in the model source tables.

Removing eight W0 and two W12 reference-inconsistent visits from training, while retaining the original folds and evaluation outcomes, gave sensitivity MAEs of 3.19/1.15/1.06. This improved follow-up error but did not establish molecular added accuracy over the identically refitted BSA-area control. The frozen primary equation and primary estimates were retained (Supplementary Table S4). Independent restricted-panel models for PP-NRS, adult DLQI and recorded ordinal IGA were less accurate. W12/W24 MAEs were 2.78/2.53 for PP-NRS and 5.90/6.28 for DLQI; exact IGA agreement was 35.4%/28.0% (Supplementary Methods).

3.5 Area and symptom inputs accounted for most predictive accuracy

Matched input ablation used the same 151/89/75 observations at W0/W12/W24 for every comparison (Fig. 5a–c). Adding BSA to the three serum markers reduced MAE from 7.67/7.48/8.56 to 3.20/1.29/1.19 points. Adding current PP-NRS further reduced follow-up error to 1.17 and 1.04 points; paired MAE differences were −0.127 (95% CI, −0.236 to −0.011) and −0.145 (−0.246 to −0.047). The matched BSA-plus-PP-NRS controls gave 1.10 and 0.96 points, so this improvement was attributable to symptom information rather than an established serum advantage.

Three markers did not lower MAE relative to CCL17 alone. With the area structure held constant, adding three markers to BSA reduced follow-up MAE by only 0.031 and 0.015 points, with both intervals spanning zero. In the separate adult DLQI subset, adding the serum panel to BSA, PP-NRS and DLQI reduced W24 MAE by 0.072 points (44 patients; 95% CI, 0.021–0.124), while increasing baseline error. These exploratory input comparisons did not establish that all three markers were necessary or optimal (Supplementary Tables S6–S7). The frozen equation was retained; the expanded-input models were not substituted for it.

Further molecular-only analyses tested nonlinear relationships, within-person changes during training and an auxiliary ordinal IGA target, while keeping deployment inputs limited to serum measurements. On the same core patients, nested selection among these candidates gave three-marker W0/W12/W24 MAEs of 7.50/7.21/7.80. A 19-marker analysis gave 7.41/7.26/8.49. These results did not approach the follow-up accuracy of E-EASI; the expanded panel did not provide a consistent improvement (Supplementary Table S14). The primary equation was unchanged.

3.6 Monitoring precision depended on the interval being assessed

Estimated changes tracked the ordering of observed improvement, with different absolute errors across intervals (Supplementary Fig. S20). Change MAEs were 3.87 for W0–W12 (85 patients; 95% CI, 3.08–4.70), 3.47 for W0–W24 (67 patients; 2.68–4.39) and 1.49 for W12–W24 (55 patients; 1.00–2.19). Corresponding Spearman correlations were 0.873, 0.940 and 0.900. The two intervals starting at baseline retained greater error despite their high correlations.

Using estimated scores at both visits to identify EASI75 (Fig. 5e) gave sensitivities of 76.5% at W12 and 86.1% at W24, with conservative conditional 95% bounds of 60.3–88.6% and 72.2–94.8%. Specificities were 98.6% (86.0–100%) and 96.2% (65.7–100%), based on 36 and 13 observed nonresponders. The broad bounds reflect the limited nonresponder samples; they combine repetition-specific binomial intervals, not independent copies of the patients. Sensitivity for EASI90 was lower, at 63.0% and 52.9% (Supplementary Fig. S21). These paired evaluations support examining response alongside the absolute reading, while preserving the distinction between monitoring agreement and clinical interchangeability.

3.7 Future-response prediction remained exploratory

Baseline CCL17, CCL26 and IL-5 gave a W12 itch-improvement AUROC of 0.630 in 102 patients. Adding them to baseline clinical information changed AUROC by 0.026 (95% CI, −0.021 to 0.074). Training-selected compact panels had higher internal discrimination, but performance fell when entire assay-report sources were held out. In a separate 68-patient analysis, W12 concentrations of the original three markers did not improve prediction of W24 EASI75 over clinical information available through W12. These analyses concern future response rather than contemporaneous EASI estimation (Supplementary Tables S18–S20).

3.8 Molecular-only transfer did not establish an advantage over p-EASI

In the published Chinese OMIX002850 dataset, an existing three-marker molecular-only equation was applied without refitting. Its W0/W16 MAEs were 11.51/6.92, compared with 11.27/3.89 for unchanged p-EASI, on the same 125/60 patients. The W16 MAE difference was 3.03 points (95% CI, 1.65–4.38), despite a higher Spearman correlation for the three-marker equation (0.557 versus 0.364). The local matched analysis likewise did not establish molecular-only superiority (Supplementary Table S15). OMIX002850 lacks BSA and therefore cannot validate E-EASI. This retrospective formula-transfer comparison was also limited by previous exploration of the dataset, unexcluded participant overlap and unbridged assay scales; it was not independent external validation.

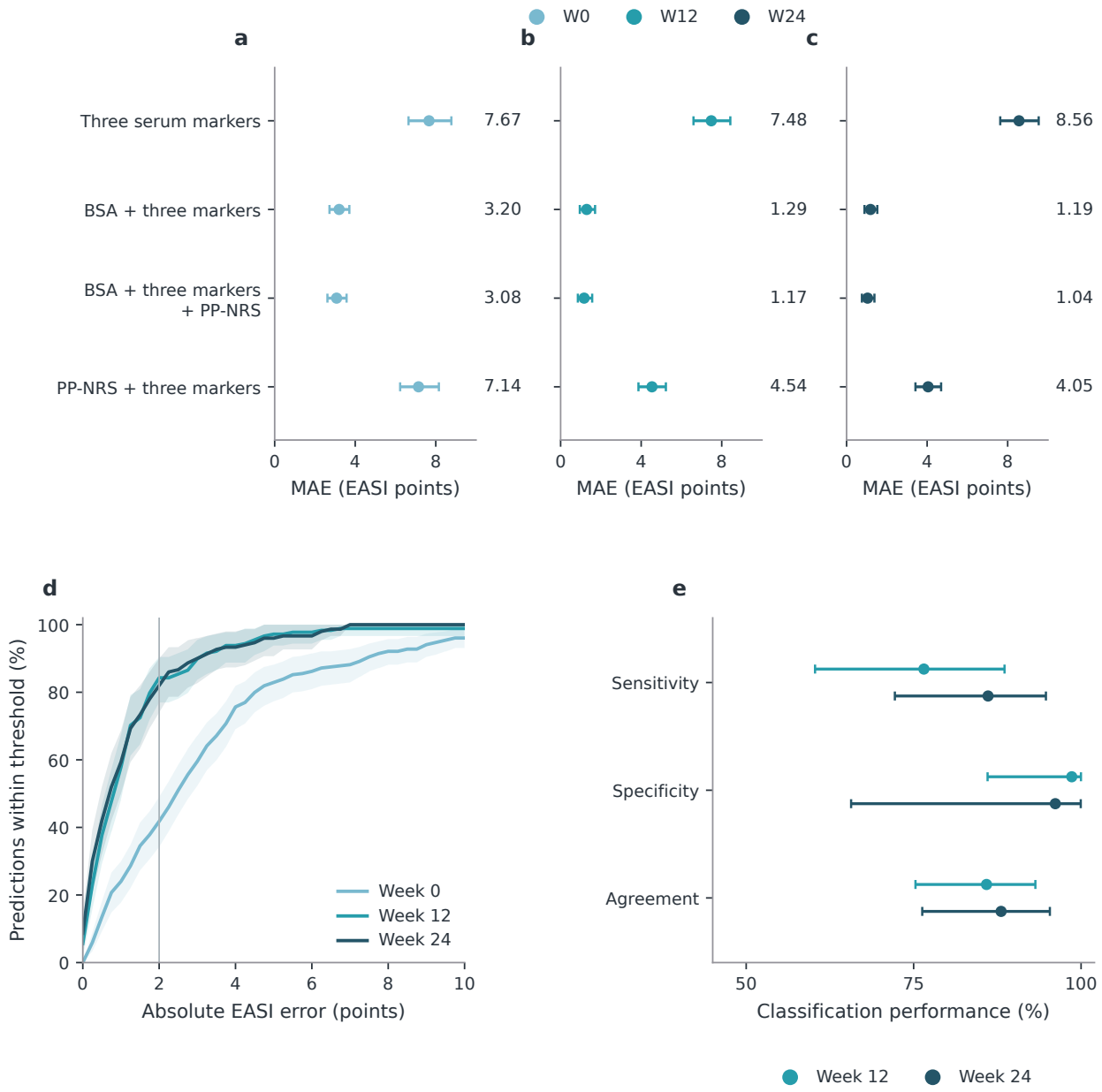


Figure 5: **Input contributions and performance of the common EASI estimator.** a–c, Post hoc input-set comparisons on the same 151/89/75 patients at W0/W12/W24. Points and labels show mean absolute error (MAE) with 95% paired-patient bootstrap intervals. Markers are CCL17, CCL26 and IL-5; clinical inputs come from the current visit. Models use the same outer patient folds, W0/W12 training, and training-only preprocessing and inner selection. The four displayed input sets contain serum markers; complete clinical-only controls and identical-input molecular comparisons remain in Table 2 and Tables S6–S7. Molecular added accuracy requires comparison at identical clinical inputs. d, The unchanged primary procedure on its original 152/89/75 patients: cumulative absolute-error proportions with pointwise bootstrap 95% intervals. W0/W12 predictions are patient-held-out. Only one W12 patient and no W24 patients have observed EASI >21. All errors enter the calculation; thresholds up to ten points are displayed. Two points is a visual reference, not an accepted interchangeability margin. e, EASI75 classification uses observed and estimated scores with their own baselines. Sensitivity uses 49/54 responders, specificity 36/13 nonresponders and agreement 85/67 paired patients at W12/W24. Conservative conditional 95% bounds average repetition-specific Clopper–Pearson limits with Bonferroni adjustment across repetitions within each endpoint, visit and metric, not all displayed metrics. The binomial working model permits dependent repetitions but does not propagate shared fitted models or development uncertainty. Metrics are scored per repetition before averaging; intervals condition on saved predictions. W24 is previously inspected same-cohort application. Separate Wilson intervals and counts are in Supplementary Table S11; additional input comparisons, CCL17 controls and marker contributions are in Tables S6–S7. Panel key: d, Individual absolute error; e, EASI75 classification.

4 Discussion

E-EASI combined affected skin area and three serum measurements in one equation across this Chinese treatment cohort. Its strongest performance was in follow-up disease activity, where most estimates were within two EASI points. Using the same calculation across visits also allowed changes to be expressed directly in EASI units. This makes the formula straightforward to examine and reproduce: the research equation takes BSA and three serum concentrations without a treatment label or previous score. Same-day clinical/serum matching was not verified, so its use as a contemporaneous clinical measurement requires a defined sampling window.

BSA is an established extent component of EASI and supplied most predictive accuracy in the ablation.[13, 14] Adding CCL17, CCL26 and IL-5 made E-EASI a clinical–molecular measure, but did not establish that three assays were necessary or that the estimate was observer-independent. Molecular trajectories address a complementary biological question. Within-group changes depended on whether all available pairs or only complete three-visit patients were analysed. IL-5 change also accompanied itch change in one group. Such associations do not by themselves demonstrate predictive benefit, and the equation's conditional coefficients are not mechanistic weights. Nonlinear molecular-only models, training on within-person change and assistance from ordinal IGA still gave large errors; expanding to 19 markers was not consistently better. This limits the present panel's use as a blood-only EASI reader without excluding future molecular approaches. Current itch modestly improved the hybrid follow-up estimate, and adult DLQI added information in a smaller subset, at the cost of additional patient-reported inputs. These findings support a practical distinction between estimating clinical severity and measuring inflammatory response.

Individual monitoring is a distinct test from agreement at a single visit. The formula reproduced much of the ordering of clinical improvement, yet changes beginning at baseline were less precise than later follow-up changes. Higher observed baseline severity was associated with larger errors. Severity mix, prior treatment and reference quality cannot be separated as causes by these data. Lesion intensity can vary despite similar affected area. Chinese serum endotypes and recent severity-linked proteomic findings also show broader inflammatory and epithelial-associated heterogeneity than a three-marker panel can represent.[21, 22] These observations motivate testing additional lesion information in severe disease, but they do not identify the cause of individual errors or establish that more blood markers will reduce them.

The p-EASI studies provide the closest precedent for a usable serum-to-EASI calculation.[8–10] Their correlations and classification results encouraged molecular estimation; the present study adds explicit absolute errors, their distribution and within-person response checks in a local treatment cohort. Their use of serum alone differs from the present BSA–serum inputs. Measurement procedures can affect calibration, and both assay compatibility and case mix matter when moving a prediction rule between settings.[11, 12] The matched local analyses therefore do not establish algorithmic superiority or show that ethnicity explains a performance difference. The OMIX002850 transfer comparison reinforces this distinction: the molecular-only equation had a higher W16 rank correlation than p-EASI but a larger absolute error. It could order patients better without giving closer clinical readings. Because BSA was unavailable and participant independence was unconfirmed, that comparison neither validates nor refutes the full E-EASI equation.

Both initial abrocitinib groups had lower adjusted itch scores at W2. This direction is consistent with the randomized JADE DARE comparison of abrocitinib 200 mg and dupilumab, which found more early itch responders with abrocitinib.[23] That trial used a PP-NRS decrease of at least four points, whereas our early target was an absolute PP-NRS of at most four; the rates are not directly comparable. Our W24 adjusted EASI and itch differences also persisted, so later improvement in all three groups should not be described as equivalence. This cohort provides observational comparisons, not a randomized head-to-head trial.

IgE and blood eosinophils require a different interpretation from skin severity. Available-case IgE medians and paired-patient changes sometimes differed in direction, showing sensitivity to the patients contributing at each visit. The abro 200 mg direction remained sensitive to the missing-data model. Dupilumab trials have also reported transient blood eosinophil increases despite clinical efficacy.[24] That literature makes a simple monotonic-decline expectation inappropriate. The revised all-available model retains patients with incomplete follow-up and compares each treatment with its own modelled baseline, without establishing a 100-versus-200 mg IgE difference. The displayed W4 interpolation assumes a smooth path; its interval propagates fitted endpoint uncertainty but cannot describe an unobserved transient rise. It therefore remains a descriptive sensitivity rather than recovered laboratory evidence.

Treatment context matters both for interpreting biomarkers and for comparing clinical outcomes. Patients receiving the higher initial abrocitinib dose had greater starting severity; subsequent reductions were selected after clinical response. Baseline adjustment and explicit maintenance pathways therefore address different questions. This distinction is consistent with randomized abrocitinib induction/withdrawal designs and observational dupilumab tapering studies, which define eligibility for maintenance or reduction before evaluating its success.[25, 26] The present results describe adjusted initial-strategy associations and observed later management, rather than the

effect of assigning each patient an alternative dose. Differential missingness, endpoint-specific residual imbalance and response-selected follow-up limit these comparisons even after baseline adjustment. They also preserve itch and quality-of-life outcomes as complementary clinical domains; neither is adequately recovered by the same small serum–BSA panel.

The local framework keeps the research EASI estimate alongside measured skin findings, symptoms, quality of life and safety (Fig. 1). Its contribution is the linked evaluation of clinical targets and serum measurements in Chinese care, rather than a first patient-inclusive consensus. Newer AHEAD and IEC frameworks clarify the difference between a visit-specific target and sustained disease control.[6, 7] Three blood-sampling visits over 24 weeks cannot establish six months of uninterrupted complete control or twelve months of off-drug remission. It does not supply automatic escalation, reduction or stopping thresholds. Independent evaluation should test the frozen equation, assay procedure and a prespecified clinical decision together. Patient-level validation prevented a person's repeated visits from appearing in both training and testing, but the molecular panel had been selected earlier and week 24 had already been inspected in prior project analyses. The current evidence consequently supports within-cohort performance of the development procedure, with a clearly defined research formula for the next evaluation. Patient bootstrap intervals condition on saved predictions and omit model-development uncertainty; no independent validation of the single final formula has been performed. Follow-up overestimation and calibration slopes below one also argue against substituting estimates for clinical examination. Adequate numbers of active severe cases, independent clinical assessment and a prespecified tolerance for individual errors will determine whether the same simple calculation can support decisions across the full disease range.

Data and code availability

Analysis code, versioned parameters, aggregate source tables and figure sources are available at `https://github.com/yanax726/T2T`. Figures and tables can be rebuilt from aggregates; model refitting requires the declared restricted input snapshots. Patient-level clinical records, serum measurements, dates and free-text notes are not included in the public repository. Access conditions and the institutional contact for these restricted records require author confirmation.

References

[1] Hywel C. Williams et al. The HOME core outcome set for clinical trials of atopic dermatitis. *Journal of Allergy and Clinical Immunology*, 149:1899–1911, 2022. doi: 10.1016/j.jaci.2022.03.017.

[2] Marjolein de Bruin-Weller et al. Treat-to-Target in Atopic Dermatitis: An International Consensus on a Set of Core Decision Points for Systemic Therapies. *Acta Dermato-Venereologica*, 101:adv00402, 2021. doi: 10.2340/00015555-3751.

[3] Jensen Yeung et al. Treat-to-target in the management of moderate-to-severe atopic dermatitis in adults: A canadian perspective. *Journal of the American Academy of Dermatology*, 89(2):372–375, 2023. doi: 10.1016/j.jaad.2023.01.053. URL `https://pubmed.ncbi.nlm.nih.gov/37074241/`.

[4] Zuotao Zhao and Xinghua Gao. Expert recommendations on treat-to-target in the systemic treatment of moderate to severe atopic dermatitis. *Chinese Journal of Dermatovenereology*, 36(8):855–864, 2022. doi: 10.13735/j.cjdv.1001-7089.202205115. In Chinese; thresholds cross-checked against the authors' 2026 clinical-trial analysis.

[5] Zuotao Zhao, Xing-Hua Gao, Yangfeng Ding, Hong Liu, Stephan Weidinger, Selena Pan, Shiqi Li, Xin Luo, Claire Feeney, Melissa Watkins, and Meena Arumugam. Efficacy of Abrocitinib and Dupilumab in Adults with Moderate-to-Severe Atopic Dermatitis Using Treat-to-Target Goals Recommended by Experts in China. *International Archives of Allergy and Immunology*, 2026. doi: 10.1159/000550965. Published online 23 February 2026.

[6] Jonathan I. Silverberg et al. Combining treat-to-target principles and shared decision-making: International expert consensus-based recommendations with a novel concept for minimal disease activity criteria in atopic dermatitis. *Journal of the European Academy of Dermatology and Venereology*, 38(11):2139–2148, 2024. doi: 10.1111/jdv.20229. URL `https://onlinelibrary.wiley.com/doi/10.1111/jdv.20229`.

[7] Joseph F. Merola et al. International Eczema Council Definitions of Low Disease Activity and Remission in Atopic Dermatitis: A Consensus Statement. *JAMA Dermatology*, 162:631–639, 2026. doi: 10.1001/jamadermatol.2026.0662.

[8] Judith L. Thijs et al. EASI p-EASI: Utilizing a combination of serum biomarkers offers an objective measurement tool for disease severity in atopic dermatitis patients. *Journal of Allergy and Clinical Immunology*, 140:1703–1705, 2017. doi: 10.1016/j.jaci.2017.06.046.

[9] Judith L. Thijs et al. EASI p-EASI: Predicting disease severity in atopic dermatitis patients treated with cyclosporin A. *Allergy*, 74:613–617, 2019. doi: 10.1111/all.13651.

[10] Daphne S. Bakker et al. EASI p-EASI: Predicting disease severity in atopic dermatitis patients treated with dupilumab using a combination of serum biomarkers. *Allergy*, 75:3287–3289, 2020. doi: 10.1111/all.14492.

[11] Kim Luijken, Laure Wynants, Maarten van Smeden, Ben Van Calster, Ewout W. Steyerberg, and Rolf H. H. Groenwold. Changing predictor measurement procedures affected the performance of prediction models in clinical examples. *Journal of Clinical Epidemiology*, 119:7–18, 2020. doi: 10.1016/j.jclinepi.2019.11.001.

[12] Richard D. Riley, Lucinda Archer, Kym I. E. Snell, Joie Ensor, Paula Dhiman, Glen P. Martin, Laura J. Bonnett, and Gary S. Collins. Evaluation of clinical prediction models (part 2): how to undertake an external validation study. *BMJ*, 384:e074820, 2024. doi: 10.1136/bmj-2023-074820.

[13] Jon M. Hanifin, Wenelia Baghoomian, Erin Grinich, Yael A. Leshem, Michael Jacobson, and Eric L. Simpson. The Eczema Area and Severity Index—A Practical Guide. *Dermatitis*, 33:187–192, 2022. doi: 10.1097/DER.0000000000000895.

[14] HOME Initiative. Eczema area and severity index: User guide, 2016. URL `https://www.homeforeczema.org/documents/easi-user-guide-dec-2016-v2.pdf`. Accessed 2026-10-17.

[15] International Council for Harmonisation. E9(r1) addendum on estimands and sensitivity analysis in clinical trials, 2019. URL `https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e9-r1-addendum-estimands-and-sensitivity-analysis-clinical-trials-guideline-statistical-principles-clinical-trials-step-5_en.pdf`. Accessed 2026-10-17.

[16] Fan Li and Fan Li. Propensity score weighting for causal inference with multiple treatments. *The Annals of Applied Statistics*, 13(4):2389–2415, 2019. doi: 10.1214/19-AOAS1282.

[17] Jonathan A. C. Sterne, Ian R. White, John B. Carlin, Michael Spratt, Patrick Royston, Michael G. Kenward, Angela M. Wood, and James R. Carpenter. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. *BMJ*, 338:b2393, 2009. doi: 10.1136/bmj.b2393.

[18] Rushani Wijesuriya, Margarita Moreno-Betancur, John B. Carlin, Ian R. White, Matteo Quartagno, and Katherine J. Lee. Multiple imputation for longitudinal data: A tutorial. *Statistics in Medicine*, 44(3-4):e10274, 2025. doi: 10.1002/sim.10274.

[19] Erik von Elm et al. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *PLoS Medicine*, 4:e296, 2007. doi: 10.1371/journal.pmed.0040296.

[20] Gary S. Collins et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.

[21] Y. Wu et al. Serum biomarker-based endotypes of atopic dermatitis in China and prediction for efficacy of dupilumab. *British Journal of Dermatology*, 188:649–660, 2023. doi: 10.1093/bjd/ljad032.

[22] Jag S. Lally, Takeshi Yoshida, Shan Gao, Xueya Cai, Donald Y. M. Leung, Lisa A. Beck, and Anna De Benedetto. Serum-Proteomic Profiling Reveals Distinct Atopic Dermatitis Severity-Linked Signatures. *Allergy*, 2026. doi: 10.1111/all.70464.

[23] Kristian Reich et al. Efficacy and safety of abrocitinib versus dupilumab in adults with moderate-to-severe atopic dermatitis: a randomised, double-blind, multicentre phase 3 trial. *Lancet*, 400(10348):273–282, 2022. doi: 10.1016/S0140-6736(22)01199-0.

[24] Michael E. Wechsler et al. Effect of dupilumab on blood eosinophil counts in patients with asthma, chronic rhinosinusitis with nasal polyps, atopic dermatitis, or eosinophilic esophagitis. *Journal of Allergy and Clinical Immunology: In Practice*, 10(10):2695–2709, 2022. doi: 10.1016/j.jaip.2022.05.019.

[25] Andrew Blauvelt et al. Abrocitinib induction, randomized withdrawal, and retreatment in patients with moderate-to-severe atopic dermatitis: Results from the JAK1 Atopic Dermatitis Efficacy and Safety (JADE) REGIMEN phase 3 trial. *Journal of the American Academy of Dermatology*, 86:104–112, 2022. doi: 10.1016/j.jaad.2021.05.075.

[26] L. S. Spekhorst et al. Successful tapering of dupilumab in patients with atopic dermatitis with low disease activity: a large pragmatic daily practice study from the BioDay registry. *British Journal of Dermatology*, 189:327–335, 2023. doi: 10.1093/bjd/ljad159.